Additional information (Not for examination): Mathematical treatment of beats

For mathematical treatment, let us consider two sound waves having same amplitude and slightly different frequencies f_1 and f_2, superimposed on each other.

Since the sound wave (pressure wave) is a longitudinal wave, let us consider $y_1 = A\sin(\omega_1 t)$ and $y_2 = A\sin(\omega_2 t)$ to be displacements of the two waves at a point $x = 0$ with same amplitude (region having high pressures) and different angular frequencies ω_1 and ω_2, respectively. Then when they are allowed to superimpose we get the net displacement

$$y = y_1 + y_2$$

$$y = A\sin(\omega_1 t) + A\sin(\omega_2 t)$$

But

$$\omega_1 = 2\pi f_1 \text{ and } \omega_2 = 2\pi f_2$$

Then

$$y = A\sin(2\pi f_1 t) + A\sin(2\pi f_2 t)$$

Using trigonometry formula

$$\sin C + \sin D = 2\cos\left(\frac{C-D}{2}\right)\sin\left(\frac{C+D}{2}\right)$$

$$y = 2A\cos\left(2\pi\left(\frac{f_1 - f_2}{2}\right)t\right)\sin\left(2\pi\left(\frac{f_1 + f_2}{2}\right)t\right)$$

Let $$y_p = 2A\cos\left(2\pi\left(\frac{f_1 - f_2}{2}\right)t\right) \qquad (11.57)$$

and if f_1 is slightly higher value than f_2 then, $\left(\frac{f_1 - f_2}{2}\right) \ll \left(\frac{f_1 + f_2}{2}\right)$ means y_p in equation (11.57) varies very slowly when compared to $\left(\frac{f_1 + f_2}{2}\right)$. Therefore

$$y = y_p \sin(2\pi f_{avg} t) \qquad (11.58)$$

This represents a simple harmonic wave of frequency which is an arithmetic average of frequencies of the individual waves, $f_{avg} = \left(\frac{f_1 + f_2}{2}\right)$ and amplitude y_p varies with time t.

Case (A):

The resultant amplitude is maximum when y_p is maximum. Since $y_p \propto \cos\left(2\pi\left(\frac{f_1 - f_2}{2}\right)t\right)$, this means maximum amplitude occurs only when cosine takes ±1,

$$\cos\left(2\pi\left(\frac{f_1 - f_2}{2}\right)t\right) = \pm 1$$

$$\Rightarrow \quad 2\pi\left(\frac{f_1 - f_2}{2}\right)t = n\pi,$$

$$\text{or, } (f_1 - f_2)t = n$$

$$\text{or, } t = \frac{n}{(f_1 - f_2)} \qquad n = 0,1,2,3, \ldots.$$

Hence, the time interval between two successive maxima is

$$t_2 - t_1 = t_3 - t_2 = \ldots = \frac{1}{(f_1 - f_2)}; \quad n = |f_1 - f_2| = \frac{1}{|t_1 - t_2|}$$

Therefore, the number of beats produced per second is equal to the reciprocal of the time interval between two consecutive maxima i.e., $|f_1 - f_2|$.

Case (B):

The resultant amplitude is minimum i.e., it is equal to zero when y_p is minimum. Since $y_p \propto \cos\left(2\pi\left(\frac{f_1 - f_2}{2}\right)t\right)$, this means, minimum occurs only when cosine takes 0,

$$\cos\left(2\pi\left(\frac{f_1 - f_2}{2}\right)t\right) = 0,$$

$$\Rightarrow \quad 2\pi\left(\frac{f_1 - f_2}{2}\right)t = (2n+1)\frac{\pi}{2},$$

$$\Rightarrow \quad (f_1 - f_2)t = \frac{1}{2}(2n+1)$$

$$\text{or, } t = \frac{1}{2}\left(\frac{2n+1}{f_1 - f_2}\right), \text{where } f_1 \neq f_2 \quad n = 0,1,2,3,\ldots.$$

Hence, the time interval between two successive minima is

$$t_2 - t_1 = t_3 - t_2 = \ldots = \frac{1}{(f_1 - f_2)}; \quad n = |f_1 - f_2| = \frac{1}{|t_1 - t_2|}$$

Therefore, the number of beats produced per second is equal to the reciprocal of the time interval between two consecutive minima i.e., $|f_1 - f_2|$.

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